

Packet Tracer - Identify MAC and IP Addresses

Objectives

Part 1: Gather PDU Information for Local Network Communication

Part 2: Gather PDU Information for Remote Network Communication

Background

This activity is optimized for viewing PDUs. The devices are already configured. You will gather PDU information in simulation mode and answer a series of questions about the data you collect.

Instructions

Part 1: Gather PDU Information for Local Network Communication

Note: Review the Reflection Questions in Part 3 before proceeding with Part 1. It will give you an idea of the type of information you will need to gather PDU information as a packet travels from 172.16.31.5 to 172.16.31.2.

- Click **172.16.31.5** and open the **Command Prompt**.
- Enter the **ping 172.16.31.2** command.
- Switch to simulation mode and repeat the **ping 172.16.31.2** command. A PDU appears next to **172.16.31.5**.
- Click the PDU and note the following information from the **OSI Model** and **Outbound PDU Layer** tabs:
 - Destination MAC Address: **000C:85CC:1DA7**
 - Source MAC Address: **00D0:D311:C788**
 - Source IP Address: **172.16.31.5**
 - Destination IP Address: **172.16.31.2**
 - At Device: **172.16.31.5**
- Click **Capture / Forward (the right arrow followed by a vertical bar)** to move the PDU to the next device. Gather the same information from Step 1d. Repeat this process until the PDU reaches its destination. Record the PDU information you gathered into a spreadsheet using a format like the table shown below:

Example Spreadsheet Format

At Device	Dest. MAC	Src MAC	Src IPv4	Dest IPv4
172.16.31.5	000C:85CC:1DA7	00D0:D311:C788	172.16.31.5	172.16.31.2
Switch1	000C:85CC:1DA7	00D0:D311:C788	N/A	N/A
Hub	N/A	N/A	N/A	N/A
172.16.31.2	00D0:D311:C788	000C:85CC:1DA7	172.16.31.2	172.16.31.5

Step 2: Gather additional PDU information from other pings.

Repeat the process in Step 1 and gather the information for the following tests:

- Ping 172.16.31.2 from 172.16.31.3.

Packet Tracer - Identify MAC and IP Addresses

- Ping 172.16.31.4 from 172.16.31.5.

Return to Realtime mode.

At Device	Dest MAC	Src MAC	Src IPv4	Dest IPv4
172.16.31.5	000C.85CC.1DA7	00D0.D311.C788	172.16.31.5	172.16.31.2
Switch1	000C.85CC.1DA7	00D0.D311.C788	N/A	N/A
Hub	N/A	N/A	N/A	N/A
172.16.31.5	00D0.D311.C788	000C.85CC.1DA7	172.16.31.2	172.16.31.5
172.16.31.3	000C.85CC.1DA7	0060.7036.2849	172.16.31.3	172.16.31.2
Hub	N/A	N/A	N/A	N/A
172.16.31.2	0060.7036.2849	000C.85CC.1DA7	172.16.31.2	172.16.31.3
172.16.31.3	0060.7036.2849	000C.85CC.1DA7	172.16.31.2	172.16.31.3
172.16.31.5	000C.CF0B.FC80	00D0.D311.C788	172.16.31.5	172.16.31.4
Switch1	000C.CF0B.FC80	00D0.D311.C788	N/A	N/A
172.16.31.4	00D0.D311.C788	000C.CF0B.FC80	172.16.31.4	172.16.31.5
172.16.31.5	00D0.D311.C788	000C.CF0B.FC80	172.16.31.4	172.16.31.5

Note: There were some stops on the way back from the original destination IP to the original src IP. The example provided does not show the hops back to the original src IP, so I did not include my findings either.

Part 2: Gather PDU Information for Remote Network Communication

In order to communicate with remote networks, a gateway device is necessary. Study the process that takes place to communicate with devices on the remote network. Pay close attention to the MAC addresses used.

Step 1: Gather PDU information as a packet travels from 172.16.31.5 to 10.10.10.2.

- Click **172.16.31.5** and open the **Command Prompt**.
- Enter the **ping 10.10.10.2** command.
- Switch to simulation mode and repeat the **ping 10.10.10.2** command. A PDU appears next to **172.16.31.5**.
- Click the PDU and note the following information from the **Outbound PDU Layer** tab:
 - Destination MAC Address: 00D0:BA8E:741A
 - Source MAC Address: 00D0:D311:C788
 - Source IP Address: 172.16.31.5
 - Destination IP Address: 10.10.10.2
 - At Device: 172.16.31.5

At Device	Dest MAC	Src MAC	Src IPv4	Dest IPv4
172.16.31.5	00D0.BA8E.741A	00D0.D311.C788	172.16.31.5	10.10.10.2
Switch1	00D0.BA8E.741A	00D0.D311.C788	N/A	N/A
Router	0060.2F84.4AB6	00D0.588C.2401	172.16.31.5	10.10.10.2
Switch0	0060.2F84.4AB6	00D0.588C.2401	N/A	N/A
Access Point	N/A	N/A	N/A	N/A
10.10.10.2	00D0.588C.2401	0060.2F84.4AB6	10.10.10.2	172.16.31.5

What device has the destination MAC that is shown?

The router.

- Click **Capture / Forward (the right arrow followed by a vertical bar)** to move the PDU to the next device. Gather the same information from Step 1d. Repeat this process until the PDU reaches its destination. Record the PDU information you gathered from pinging 172.16.31.5 to 10.10.10.2 into a spreadsheet using a format like the sample table shown below:

At Device	Dest. MAC	Src MAC	Src IPv4	Dest IPv4
172.16.31.5	00D0:BA8E:741A	00D0:D311:C788	172.16.31.5	10.10.10.2
Switch1	00D0:BA8E:741A	00D0:D311:C788	N/A	N/A
Router	0060:2F84:4AB6	00D0:588C:2401	172.16.31.5	10.10.10.2
Switch0	0060:2F84:4AB6	00D0:588C:2401	N/A	N/A
Access Point	N/A	N/A	N/A	N/A
10.10.10.2	00D0:588C:2401	0060:2F84:4AB6	10.10.10.2	172.16.31.5

Reflection Questions

Answer the following questions regarding the captured data:

- Were there different types of cables/media used to connect devices?
Yes, there are several different types of cables/media used to connect the devices.
- Did the cables change the handling of the PDU in any way?
Not that I noticed between any of the different sets of pings I did.
- Did the **Hub** lose any of the information that it received?
No, but there is no MAC address or IP for the hub. It simply passed on the information.

4. What does the **Hub** do with MAC addresses and IP addresses?

It doesn't seem like it did anything but keep passing on the information.

5. Did the wireless **Access Point** do anything with the information given to it?

A wireless Access point connects wired devices to wireless devices, so it encapsulates the packet so that the wireless devices can handle/receive it.

6. Was any MAC or IP address lost during the wireless transfer?

No MAC or IP address was lost during the wireless transfer.

7. What was the highest OSI layer that the **Hub** and **Access Point** used?

Because there was no MAC address or IP address for the Hub or Access Point, we can assume they were not using the data link layer, or the network layer, so therefore the highest layer they used was the physical layer; layer 1.

8. Did the **Hub** or **Access Point** ever replicate a PDU that was rejected with a red "X"?

Yes, sometimes it would resend the packet.

9. When examining the **PDU Details** tab, which MAC address appeared first, the source or the destination?

The destination.

10. Why would the MAC addresses appear in this order?

So that packets can be sent faster. In instances where cut-through switching is used, only the destination address is needed to send the frame forward without checking the rest of it for errors.

11. Was there a pattern to the MAC addressing in the simulation?

Only that the destination address would switch with the src address when sending the reply back.

12. Did the switches ever replicate a PDU that was rejected with a red "X"?

No, they only pass along the package.

13. Every time that the PDU was sent between the 10 network and the 172 network, there was a point where the MAC addresses suddenly changed. Where did that occur?

At the router.

14. Which device uses MAC addresses that start with 00D0:BA?

The router.

15. What devices did the other MAC addresses belong to?

The device with IP 10.10.10.2, the device with IP 172.16.31.5, Switch1, and Switch0.

16. Did the sending and receiving IPv4 addresses change fields in any of the PDUs?

Only when sending the reply back, the IP addresses would switch, other than that, no.

17. When you follow the reply to a ping, sometimes called a *pong*, do you see the sending and receiving IPv4 addresses switch?

Yes, we see them switch at the very end when receiving the reply.

18. What is the pattern to the IPv4 addressing used in this simulation?

The IP addresses in a specific group share similar but different addresses. (Two groups: 10.10.10.x, and 172.16.31.y, x and y being varying digits for the last portion of the IP)

19. Why do different IP networks need to be assigned to different ports of a router?

So that the different networks can be connected.

20. If this simulation was configured with IPv6 instead of IPv4, what would be different?

IPv6 might be faster, but nothing else.